$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 3\,k^2\,\overset{(4)}{\kappa}_{15} + 3\,\overset{(2)}{\kappa}_1
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \mathrm{Y}_1\,k^2 + 2\,\overset{(2)}{\kappa}_1 & \frac{\mathrm{Y}_1\,k^2+2\,\overset{(2)}{\kappa}_1}{\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\mathrm{Y}_1\,k^2+2\,\overset{(2)}{\kappa}_1}{\sqrt{2}} & \frac{1}{2}\,(\mathrm{Y}_1\,k^2 + 2\,\overset{(2)}{\kappa}_1) & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & k^2\,\overset{(4)}{\kappa}_1 & -\frac{k^2\,\overset{(4)}{\kappa}_1}{\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{k^2\,\overset{(4)}{\kappa}_1}{\sqrt{2}} & \frac{k^2\,\overset{(4)}{\kappa}_1}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{3\,(k^2\,\overset{(4)}{\kappa}_{15} + \overset{(2)}{\kappa}_1)}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{2}{3\,\mathrm{Det}\,(1^+)} & \frac{\sqrt{2}}{3\,\mathrm{Det}\,(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\sqrt{2}}{3\,\mathrm{Det}\,(1^+)} & \frac{1}{3\,\mathrm{Det}\,(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{4}{9\,k^2\,\overset{(4)}{\kappa}_1} & -\frac{2\,\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_1} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{2\,\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_1} & \frac{2}{9\,k^2\,\overset{(4)}{\kappa}_1}
\end{array}
$$

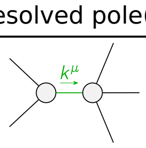
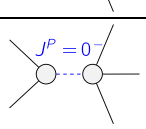
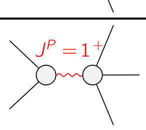
$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

Abbreviations used in matrices

$$\mathrm{Y}_1 == 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 \;\&\&\; \mathrm{Det}\,(1^+) == k^2\,(3\,\overset{(4)}{\kappa}_{15} - \frac{3\,\overset{(4)}{\kappa}_3}{2}) + 3\,\overset{(2)}{\kappa}_1$$

Lagrangian

$$\overset{(2)}{\kappa}_1\,\mathcal{K}_{\alpha\beta\chi}\,\mathcal{K}^{\alpha\beta\chi} - 2\,\overset{(2)}{\kappa}_1\,\mathcal{K}^{\alpha\beta\chi}\,\mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} - \overset{(4)}{\kappa}_1\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - 2\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\,\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$	
$\mathcal{J}^{\#1\,\alpha}_{1^-} + 2\,\mathcal{J}^{\#2\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == \mathcal{J}^{\#2\,\alpha\beta}_{1^+}$	3	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$	
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	17		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{\overset{(4)}{\kappa}_1}$
	1	$-\frac{\overset{(2)}{\kappa}_1}{\overset{(4)}{\kappa}_{15}}$	$-\frac{1}{3\,\overset{(4)}{\kappa}_{15}}$
	3	$\frac{2\,\overset{(2)}{\kappa}_1}{-2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_3}$	$\frac{2}{6\,\overset{(4)}{\kappa}_{15} - 3\,\overset{(4)}{\kappa}_3}$
Resolved unitarity condition(s):	(Demonstrably impossible)		